

Tae Yong Song

Ph.D. Student in Building Science

Building Simulation Lab (BS Lab)

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EDUCATION

- 2026.03–present** **Ph.D. in Architecture and Architectural Engineering**, Seoul National University
Building Simulation Lab (BS Lab); Advisor: Professor Cheol-Soo Park
- 2022.03–2024.02** **M.S. in Architecture and Architectural Engineering**, Seoul National University
Thesis: “Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model”
- 2015.03–2022.02** **B.S. in Architecture and Architectural Engineering**, Seoul National University

PUBLICATIONS

International Journal Article

- 2025** Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288. Published online 5 August 2025; issue dated 2026.

International Conference Papers

- 2026** Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Tae Yong Song, Jin-Hong Kim, Seung-Ju Lee, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.
- 2024** Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.
- 2023** Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.
- 2023** Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

- 2026** Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms

Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Outstanding Presentation Award.**
- 2022** Dynamic Sensitivity Analysis Using NODE-GAM

Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

2026.03–2030.12	Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation <i>Korea Institute of Civil Engineering and Building Technology (KICT)</i>
2025.10–2026.04	LLM Technology for Digital-Twin-Based Building Energy Models <i>Samsung Electronics</i>
2025.12–2026.01	IDF Model Development for Digital-Twin Energy Simulation <i>Eterion (Samsung Electronics)</i>
2022.04–2024.12	Automated Digital Diagnosis and Design Technologies for Green Remodeling <i>Korea Agency for Infrastructure Technology Advancement (KAITA)</i>
2023.05–2023.11	Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control <i>LG Electronics</i>
2022.12–2023.07	Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling <i>Ministry of Land, Infrastructure and Transport (MOLIT)</i>

TEACHING EXPERIENCE

2026–present	Teaching Assistant , Seoul National University Department of Architecture and Architectural Engineering
2023	Teaching Assistant , Seoul National University Department of Architecture and Architectural Engineering

HONOR

2026	Outstanding Presentation Award Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division
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